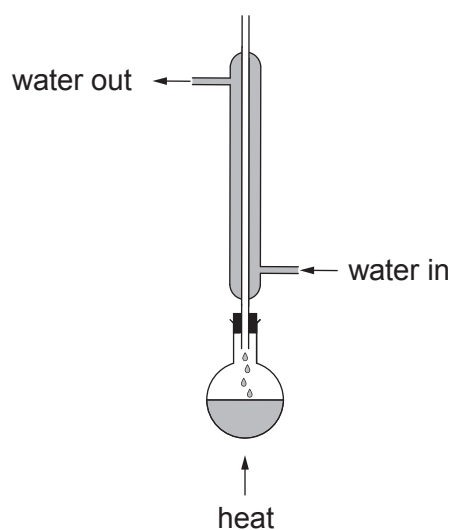


9. (a) Alcohols react with carboxylic acids to form esters. To prepare a pure sample of an ester a condenser can be used in the first stage.



- (i) Name the method being used with the condenser positioned in this way. Explain why it is necessary to use the condenser. [2]

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.....

.....

- (ii) Name the method used to separate the product from the liquid mixture. State which property of an ester allows this method of separation to be used. [2]

Method

Property

- (iii) Name the catalyst most commonly used in the preparation of an ester. [1]

.....

- (iv) Write the equation for the reaction between methanoic acid and butan-2-ol. Clearly show the structure of the ester formed. [2]



- (b) A compound is known to be one of butan-2-ol, $\text{CH}_3\text{CH}_2\text{CH}(\text{OH})\text{CH}_3$, 2-methylpropanoic acid, $\text{CH}_3\text{CH}(\text{CH}_3)\text{COOH}$, and 3-hydroxybutanoic acid, $\text{CH}_3\text{CH}(\text{OH})\text{CH}_2\text{COOH}$.

Choose **two** chemical tests that will enable you to determine which one it is.

Complete the table below.

Give the reagent(s) for each test and the observation expected for a positive result.

For each test put a **tick (✓)** in the box to show the compound that gives a positive result and a **cross (✗)** for those that do not. [5]

Reagent(s)	Observation expected for positive result	butan-2-ol	2-methylpropanoic acid	3-hydroxybutanoic acid

